

A Mechanism and Simple Model of the North Atlantic Oscillation and Annular Modes

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(6 January 2003)

ABSTRACT

We present a model for the basic spatial and temporal structure of the large-scale modes of intraseasonal variability and associated variations in the zonal index. Such variability in the extra-tropical atmosphere is known to be organized into fairly well-defined patterns or modes, and among the most prominent are the North Atlantic Oscillation (NAO) and a more zonally symmetric pattern known as an annular mode, most pronounced in the Southern Hemisphere. We suggest that their structure is a direct consequence of the stirring effects of baroclinic eddies and we explicitly show how such stirring, as represented by a simple random forcing in a barotropic model, leads to a variability in the zonal flow via a variability in the eddy momentum flux convergence. Typically, the leading modes of variability may be characterized as ‘wobbles’ in the zonal jet position and ‘pulses’ in the zonal jet strength. For a large parameter regime these give rise to patterns of low-frequency variability that resemble those observed. If the stochastic forcing is statistically zonally uniform, then the resulting patterns of variability (i.e., the empirical orthogonal functions) are zonally uniform and the pressure pattern is dipolar in the meridional direction, resembling an annular mode. If the forcing is enhanced in a zonally localized region, thus mimicking the effects of a stormtrack over the ocean, then the resulting variability pattern is zonally localized, with a pattern resembling the North Atlantic Oscillation. This suggests that the North Atlantic Oscillation and annular modes are produced by the same mechanism, and are manifestations of the same phenomenon.

The timescale of variability of the patterns is longer than the decorrelation timescale of the stochastic forcing, being reddened by nonlinear dynamics and by the linear effects of friction. For reasonable parameters this produces decorrelation time of order 10 days. The model also produces some long term (e.g., 100–1000 days) variability of the phenomena, without imposing such variability via the external parameters except in the nearly white stochastic forcing. One consequence of this picture is that any oceanic influence that might produce variability on these structures on these and even longer timescales is likely to be through its effect on baroclinic activity, rather than directly on the large-scale patterns themselves.

1. Introduction

The large-scale atmospheric circulation displays variability on multiple timescales. The most prominent variability in the extratropics is that due to baroclinic eddies, or midlatitude weather systems, which typically have a timescale of a few days. On timescales of a season or longer atmospheric variability may be influenced by interactions at its boundaries, provided in part by the ocean, and other slow changes in forcing. The variability on intermediate (i.e., intraseasonal) timescales, say between 10 days and a season (and often confusingly called ‘low-frequency’ variability) has a less obvious cause. The direct effect of the ocean or other changing boundary seems unlikely to be important, both because large-scale sea-surface temperatures tend to change primarily on still longer timescales and because the effect is unlikely to be strong enough to produce discernible changes in the atmospheric circulation on the 10–100 day timescale. Rather, we expect that this variability has a primarily atmospheric origin, perhaps involving planetary waves or arising from baroclinic activity and weather systems and reddened by frictional or nonlinear processes. The dynamics of such variability, however, not fully understood, and this is the subject of this paper.

There are two aspects of such intraseasonal variability: its timescales and its spatial structure. There is no pronounced peak in the power spectrum of the atmospheric fields on the weeks-to-months timescale, but nor is there a dip in the spectrum at timescales longer than that associated with baroclinic eddies. This suggests that the variability at intraseasonal timescales may be, at leading order, caused by a reddening of the power spectrum of the known forcing (i.e., baroclinic instability) by frictional processes and/or the nonlinear dynamical effect of a cascade of variance to larger spatial scales and longer frequencies. At still longer timescales, there may be additional power on the year-to-decade timescale, and whether this can have a purely atmospheric origin is not known.

If the atmospheric fields are appropriately filtered in time to select intraseasonal timescales, then fairly well-defined patterns of variability emerge. This spatial structure has been the object of much study and debate, going back at least as far as Walker and Bliss (1932) and summarized recently by Wallace (2000) and Wanner *et al.* (2002). These patterns robustly show up in correlation maps or teleconnection patterns (Wallace and Gutzler 1981), and in Empirical Orthogonal Function (EOF) analysis of the low-passed fields (e.g., Ambaum *et al.* 2001). A host of patterns with low-frequency behavior have been identified by these and other workers — the North Atlantic

Oscillation (NAO), the Northern and Southern Annular Modes (NAM, SAM), the Pacific North American pattern (PNA), the Pacific Decadal Oscillation (PDO) and so on. Of these the North Atlantic Oscillation is perhaps most well known and this, and the so-called annular modes, will mainly concern us from now on. The scale of these patterns is significantly larger than that typically associated with a single baroclinic eddy, ranging from a few thousand kilometers of the NAO to the hemispheric scale of the annular modes. Two other aspects of their structure stand out: (i) They are barotropic, or at least equivalent barotropic (little phase shift in the vertical); (ii) There is a strong dipolar component in their horizontal structure, especially in the pressure field.

There are certainly many hints, both observational and theoretical, that suggest that baroclinic eddy activity (i.e., weather systems) is largely responsible for producing such patterns, even though the patterns are much larger than that of a single baroclinic system. Large-scale eddy-driven structures often tend to be barotropic because the life-cycle of baroclinic eddies is characterized by a barotropic decay and a cascade to larger horizontal *and vertical* scales (Simmons and Hoskins 1978; Salmon 1980). On the observational side, analysis of annular modes and the NAO indicates that transient, high frequency (i.e., 1–10 days) activity plays an important role in maintaining their variability (e.g., Lau 1988; Limpasuvan and Hartmann 2000; DeWeaver and Nigam 2000). Consistently, the midlatitude jet in the Atlantic sector is stronger during periods of high NAO index [Ambaum et al. (2001), their figures 6 and 7], and this jet is fairly barotropic, indicating an eddy-driven origin. Finally, recent experiments with a GCM (Cash et al. 2002) have shown a strong correlation between the location and strength of the dipole with the location and strength of the baroclinic eddy activity.

These findings all suggest that the essential dynamics are barotropic (except for the ultimate source of the stirring itself, which is baroclinic instability) and that eddy activity is important. Our goal is to understand if and how such higher-frequency (1–10 days) eddy dynamics can produce the patterns characteristic of the lower frequency (10–100 days) patterns. Specifically, we seek to present a simple dynamical model, perhaps the simplest possible dynamical model, of the phenomena (specifically the NAO and annular modes) in order to shed insight on the dynamics that give rise to the spatial patterns and the temporal variability of the structures. We shall not present a complete model or a complete theory. Rather, our model might be considered as a ‘dynamical null-hypothesis’ that might be built upon to create a more complete theory, or that might be dis-

carded if observations or extensive simulations with GCMs indicate that the proposed dynamics are fundamentally incorrect.

2. The Model

a. Jets on a β -plane

Consider first the maintenance of the extratropical jet. This jet has a different dynamical origin from the highly baroclinic subtropical jet. The latter arises from a thermal wind balance with the strong meridional temperature gradients at the edge of the Hadley Cell, whereas the former is driven by eddy momentum flux convergence in midlatitude weather systems and, because these largely occur in the mature phase of the baroclinic lifecycle, they act to produce a predominantly barotropic jet. In reality the subtropical and midlatitude jet are often not geographically distinct because the polar limit of the Hadley cell overlaps the equatorial limit of the midlatitude baroclinic zone, and the jets often appear as one. However, there are two distinct mechanisms at work.

A simple barotropic model illustrates the mechanisms of the eddy driven jet (e.g., Held 2000). For two-dimensional incompressible flow the barotropic zonal momentum equation is

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} - f v = -\frac{\partial \phi}{\partial x} + F_u - D_u \quad (2.1)$$

where F_u and D_u represent the effects of any forcing and dissipation. Now, the meridional momentum and vorticity fluxes are related by the identity

$$v \zeta = \frac{1}{2} \frac{\partial}{\partial x} (v^2 - u^2) - \frac{\partial}{\partial y} (uv) \quad (2.2)$$

so that with cyclic boundary conditions

$$\overline{v' \zeta'} = -\frac{\partial \overline{u' v'}}{\partial y}. \quad (2.3)$$

where the overbar denotes a zonal average. Equation (2.3) also holds, locally in x , if the average is a time or ensemble average provided that the eddy statistics are zonally uniform.) Averaging (2.1) thus gives

$$\frac{\partial \bar{u}}{\partial t} = \overline{v' \zeta'} + \bar{F}_u - \bar{D}_u \quad (2.4)$$

using also that $\bar{v} = 0$, a result that again also holds in a time or ensemble average if the eddy statistics are zonally uniform.

Typically, there will be little direct forcing of the momentum, and if friction is parameterized by a linear drag then

$$\frac{\partial \bar{u}}{\partial t} = \overline{v'\zeta'} - r\bar{u} \quad (2.5)$$

where r is an inverse frictional timescale. Now consider the maintenance of this vorticity flux. The barotropic vorticity equation is

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + v\beta = F_\zeta - D_\zeta \quad (2.6)$$

where F_ζ parameterizes the stirring of barotropic vorticity.

Linearize about a mean zonal flow to give

$$\frac{\partial \zeta'}{\partial t} + \bar{u} \frac{\partial \zeta'}{\partial x} + \gamma v = F'_\zeta - D'_\zeta \quad (2.7)$$

where

$$\gamma = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} \quad (2.8)$$

is the meridional gradient of absolute vorticity. From (2.7) form the pseudo-momentum equation by multiplying by ζ'/γ and zonally averaging, whence

$$\frac{\partial M}{\partial t} + \overline{v'\zeta'} = \frac{1}{\gamma} (\overline{\zeta'F'_\zeta} - \overline{\zeta'D'_\zeta}) \quad (2.9)$$

where

$$M = -\frac{1}{2\gamma} \overline{\zeta'^2} \quad (2.10)$$

is the (negative of) the pseudomomentum. From (2.5) and (2.9) we obtain

$$\frac{\partial \bar{u}}{\partial t} + \frac{\partial M}{\partial t} = -r\bar{u} + \frac{1}{\gamma} (\overline{\zeta'F'_\zeta} - \overline{\zeta'D'_\zeta}), \quad (2.11)$$

and in a statistically steady state

$$r\bar{u} = \frac{1}{\gamma} (\overline{\zeta'F'_\zeta} - \overline{\zeta'D'_\zeta}). \quad (2.12)$$

As the terms on the right-hand-side represent the stirring and dissipation of vorticity, then on a β -plane, or on the surface of a rotating sphere, then meridionally localised but otherwise relatively unstructured vorticity stirring will give rise to an eastward mean zonal flow in the region of the stirring, with a westward flow north and south of the stirring region. These equations represent the well-known physical argument that stirring gives rise to Rossby wave generation, and that momentum will converge in the region of stirring as the Rossby waves propagate away and dissipate.

This mechanism seems, at least superficially, different from that often used to describe the formation of jets on a β -plane when a fluid on a β -plane is stirred spatially uniformly (see Lee 1997). In that case, multiple jets often ensue with a width determined by the intensity of the stirring and the magnitude of β and friction. The mechanism of jet formation is then often expressed in terms of an inverse energy cascade to larger scales, inhibited by the formation of Rossby waves, leading to the preferential formation of zonal flow (Vallis and Maltrud 1993). However, equations (2.11) and (2.12) are still valid in that case and the jet can still be considered to be maintained by the stirring of pseudomomentum $(\overline{\zeta'F'_\zeta})/\gamma$. The difference lies in the width of the stirring region. If the stirring region is narrow then Rossby waves can escape from the stirring region before dissipating then momentum will converge on the stirred region and there will be just one eastward with a location roughly coincident with the stirring. However, if the stirring region is sufficiently wide, wider than the natural jet scale (which is determined in part by the value of β) then multiple jets can form inside the stirring region. In the earth's atmosphere the stirring region (i.e., the midlatitude baroclinic zone) is relatively narrow in the sense that there is normally only one region of eddy driven eastward flow in the mean, However, the baroclinic zone is typically wider than the instantaneous jet itself, and the jet may thus meander within the baroclinic zone.

The stirring that might generate such jets arises from baroclinic instability or, more precisely, from the transfer of energy from baroclinic to barotropic modes. To see this, consider the two-layer quasi-geostrophic equations

$$\frac{\partial q_i}{\partial t} + J(\psi_i, q_i) = 0, \quad i = 1, 2 \quad (2.13)$$

where

$$q_i = \nabla^2 \psi_i + F(\psi_j - \psi_i) + \beta y, \quad j = 3 - i. \quad (2.14)$$

and F is the inverse square deformation radius. If this is decomposed into barotropic and baroclinic modes in the standard way, the evolution equation for the barotropic mode becomes

$$\frac{\partial}{\partial t} \nabla^2 \psi + J(\psi, \nabla^2 \psi + \beta y) = J(\tau, \nabla^2 \tau) \quad (2.15)$$

where $\psi = (\psi_1 + \psi_2)/2$ and $\tau = (\psi_1 - \psi_2)/2$. The term on the right-hand-side is just the stirring of the barotropic mode by the baroclinic mode, leading to transfer of energy into the barotropic mode as part of the baroclinic lifecycle. Such stirring by baroclinic eddies thus gives rise to momentum convergence and is the ultimate cause of the surface westerly winds in midlatitudes.

The vorticity flux producing the zonal jet will, of course, fluctuate simply because baroclinic activity fluctuates, partly in response to variations in the zonal shear itself and partly because it is a turbulent, chaotic system, and these fluctuations will give rise to variations in the zonal index (e.g., Feldstein and Lee 1998; Lorenz and Hartmann 2001). Such variations will be largely barotropic, but the presence of surface drag may lead to the generation of a shear, which in turn will produce baroclinic activity and enhanced stirring, and in this way longer timescales might be produced (Robinson 2000). Here we shall restrict ourselves to the simpler case of a barotropic model, and we will model the stirring simply by a random process with time and space scales chosen to roughly mimic those of baroclinic instability.

Suppose the stirring is statistically zonally uniform and confined to midlatitudes. The zonally averaged zonal wind will then fluctuate in response to this stirring, and we will see in Section 3 that typical fluctuations are in fact such as to produce a dipolar structure in the pressure or streamfunction field, similar to that of observed annular modes. Clearly, stronger stirring will give rise to a stronger and perhaps more variable jet, and thus we might expect a stronger and more variable jet in regions of storm tracks. However, the argument above strictly applies only to a zonally averaged zonal wind, because of the use of (2.3). Nevertheless, if the x -averaging is over a zonal distance longer than the scale of a typical eddy, or if the zonal scale over which the eddy statistics vary is longer than the meridional scale, then the first term on the right-hand-side of (2.2) will be smaller than the second term, after time averaging, and (2.3) will *approximately* hold. Similarly, in these circumstances the zonal advection of momentum in (2.1) will be smaller than that of meridional momentum, and the upshot is that (2.5) will approximately hold regionally, without the need for complete zonal averaging. The consequence of this is that if the stirring is enhanced in some longitudinally localized region then, provided the zonal extent of the stirred region is larger than the eddy scale itself, the zonal jet will be both stronger and more strongly fluctuating than average in that region. [In an oceanic context and in a laboratory experiment Rhines (pers. comm) has shown how local stirring can generate a mean flow.] The net effect will be to produce a localized dipole in the pressure or streamfunction field — the characteristic pattern of the NAO.

Regarding the temporal structure of the response: suppose that, for simplicity, the stirring is white noise. Then, because of the effects of friction in (2.5) and possibly because of the effects of the inverse cascade, the response will be redder than the stirring — that is to say, it will have

more power at lower frequencies. Thus, the posited patterns will be most apparent in the temporally low-passed fields, as this will filter the noise of the stirring itself. Eddy stirring is thus a mechanism for the production of both zonally averaged variability and zonally localized structures. To see whether such stirring can indeed produce the characteristic spatial patterns and temporal variability of the observed intraseasonal modes of variability, such as annular modes and the NAO, we turn to numerical integration.

b. Numerical model

We integrate the barotropic vorticity equation on the sphere, namely

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta + f) = S - r\zeta + \kappa \nabla^4 \zeta. \quad (2.16)$$

The notation is standard, with $f = 2\Omega \sin \vartheta$, where ϑ is latitude, ζ is vorticity and ψ stream-function. The model is spectral with the nonlinear term evaluated without aliasing using a spectral transform method. Typically, the model is run at a resolution of T42 or T84, which is adequate because our concern is large-scale patterns. The last two terms on the right hand side of (2.16) are a linear drag and a term to remove the enstrophy that cascades to small scales, being the simplest parameterizations of those processes that remove momentum and enstrophy from the flow. The coefficient κ depends on the model resolution, for that term is a subgrid-scale closure. The linear drag has some physical grounding in Ekman layer theory, and for a barotropic representation of the atmosphere we expect that r takes values of order $1/10 \text{ days}^{-1}$.

The term S represents stirring of the barotropic flow by baroclinic eddies, and we represent this by a Markov process, similar to that employed in Maltrud and Vallis (1991). Typically, we choose to excite a small range of wavenumbers, $n_{\min} < n < n_{\max}$ where n is the total wavenumber and, for example, $n_{\min} = 8$ and $n_{\max} = 12$, except that small zonal wavenumbers, including the zonal flow, are excluded from the forcing. The excitation of each wavenumber (m, n) is determined by the ordinary differential equation

$$\frac{dS_{mn}}{dt} = \dot{W}_{mn} - \alpha S_{mn} \quad (2.17)$$

where $\dot{W}$ is a white noise process (a different realisation for each wavenumber) and the parameter α determines the decorrelation time of the forcing. S_{mn} is thus an Ornstein-Uhlenbeck process. To

implement (2.17), we use the related finite difference equation,

$$S_{mn}^i = \tau W_{mn} + \sqrt{1 - \tau^2} S_{mn}^{i-1} dt \quad (2.18)$$

where dt is the model timestep, the superscript i is the timestep index, and τ is the prescribed decorrelation time of the forcing, which we typically choose to be two days.

This spectral forcing is then transformed to physical space, where it is masked such that it has a non-negligible amplitude only in midlatitudes, typically between about 40° and 60° latitude. For some experiments it is also made statistically zonally nonuniform; that is, it is enhanced in a region about 45° wide in longitude, to mimic the effects of enhanced stirring in stormtracks. Instantaneously, the forcing is generally zonally nonuniform, and in all cases is constructed to have zero projection on the zonally symmetric flow (i.e., all components with $m = 0$ are zero).

Apart from this meridional masking and the choice of the scale of the stirring, the stochastic forcing is relatively unstructured, and the resulting momentum flux convergences result from the nonlinear dynamics of the model. This type of stochastic model differs from that used in, for example, Branstator (1992) or Whitaker and Sardeshmukh (1998), in which the model is linear and the mean flow is taken from observations or a GCM. Here the model is nonlinear, and it is the stochastic forcing in conjunction with nonlinear dynamics that generates the mean flow, and that is important for its pattern of variability.

3. Spatial Structure

a. Mean state for zonally symmetric model

A typical time and zonally averaged zonal wind, and the rms (i.e., eddy) velocity are illustrated in fig. 1. The stochastic forcing is a Gaussian centered at 45° with a standard deviation of 12° , and a decorrelation timescale of two days. A strong westward jet emerges in the region of the forcing, flanked by two eastward jets, rather stronger on the equatorial side indicating enhanced breaking on the equatorial side. Consistently, the mean position of the jet is somewhat *poleward* of the center of the stirring, and this polewards offset increases slightly as the forcing strength increases. The eddy velocities are of the same magnitude, albeit a little larger than, the zonally averaged velocity, a characteristic also of the flow in the earth's atmosphere. This is a consequence of the rather limited inverse cascade, in both model and atmosphere, which limits the magnitude of the eddies.

The natural meridional scale of a jet in homogeneous barotropic turbulence is determined by the eddy kinetic energy, and the value of β and friction (e.g., Vallis and Maltrud 1993). As mentioned in section 2a, if the meridional extent of the forcing region is allowed to become larger than that jet scale, multiple jets may form within the forcing region. This phenomena is illustrated in fig. 2. Interestingly, the transitions from one jet regime to another are not smooth; for example, as the forcing region broadens sufficiently there is a sharp transition from one jet to two, and the locations of the ensuing two jets are both quite different from that of the single jet.

b. Variability

Let us now examine the variability of a single, eddy-driven jet. Consider the momentum equation (2.5) and suppose that the vorticity flux is such as to produce an eastward jet in midlatitudes, and that its magnitude fluctuates temporally but that its meridional structure remains fixed. Then the zonally averaged zonal wind will fluctuate in place — it will pulse — and the associated EOF of the zonal wind will have a tripole structure (fig. 3). Since at each instant the pressure field (the streamfunction) and the velocity are linearly related the associated variability in the pressure field can be expected to be a dipole. If the zonal wind fluctuates in this way, the node of the pressure EOF will coincide with the maximum of the jet, which in turn occurs where the stirring is strongest.

The other dominant mode of variability might be termed a wobbling of the zonal jet, that is an oscillation in its mean latitude without necessarily any change in amplitude (fig. 3). Both pulsing and wobbling behaviour frequently occur in numerical simulations (see fig. 4 and fig. 5). Typically, the first two EOFs of the zonally symmetric model represent either pulses or wobbles, or a combination thereof. The order that these two modes appear seems somewhat sensitive to the nature of the forcing used, but they are nearly always the first two modes of the zonally averaged zonal velocity. Typically we find that the wobbling EOF is dominant when the zonal scale of the stochastic forcing is around wavenumber 10 or smaller, and in that case the second EOF re when the stochastic forcing has a larger scale, the first EOF tends to have a more mixed character, as in fig. 5. The meridional structure of a pure pulsing EOF more-or-less mirrors that of the jet itself with a tripolar structure in the zonal wind, and the wobbling mode is in quadrature with this. Note that in both cases (see fig. ??b and fig. 5b) the first EOF of the streamfunction is dipolar, with a node somewhat poleward of the mean position of the jet and the lower band more or less coincident with the mean position of the jet. These

structures are apparent in both the EOF of the zonally averaged fields, and in the zonally averaged EOF of the two dimensional fields. In the former case the variance accounted for by the first two EOFs is typically over 30% each and these are both well separated from the other EOFs. Similar structures are seen in the observations (Feldstein and Lee 1998; Lorenz and Hartmann 2001) and in simulations with a general circulation model (Cash et al. 2002). Indeed Feldstein and Lee (1998) characterizes the EOFs of the northern and southern hemispheres as being either a strengthening and weakening of the jet, or a latitudinal movement of the jet, although the interpretation is complicated by additional variations in the subtropical jet. Although very descriptive, the pulsing and wobbling modes are not wholly independent. Recall that the mean wind was somewhat poleward of the center of the stirring, because of predominantly equatorward breaking of the Rossby waves. If the stirring is stronger, the jet is not only stronger but is pushed polewards, and the pulse and the wobble are synchronized. The EOFs are describing this in the most economical way possible, subject to their orthogonality.

When one looks at the EOF of the two-dimensional fields, the zonal average of the first EOF (of either velocity or streamfunction) is usually very similar to the EOF of the zonally average field, although the subsequent EOFs are less distinct. The first EOF normally is well separated from the others, although the variance accounted for is typically less than 20%. The first EOF of the two-dimensional streamfunction is illustrated in fig. 6. It is nearly zonally symmetric and may be taken as a definition of the annular mode of this model. Any zonal asymmetry here can be ascribed to sampling issues, that is, a finite length of numerical integration. However, although the leading EOF is not guaranteed to be zonally symmetric, its existence should not lead one to necessarily conclude that there is a strong mode of hemispheric-wide variability in the model. The EOF analysis is again merely seeking the most economical description of model variability. In particular, in most of the integrations we have examined the zonal flow does not vary synchronously across the hemisphere. The one-point correlation function shows this quantitatively (fig. 7). This figure shows the one-point correlation maps for various base points. (The base points can be easily determined from the figure as the correlation is one at the base point.) In the meridional direction, the dipolar structure of the EOF can be seen in the correlation function — there is a negative correlation across the dipole. The zonal scale of the correlation is related to the scale of the energy containing eddies, as one might expect given that the spatial correlation function is essentially the Fourier transform of the variance

of that variable (so the velocity correlation function is the Fourier transform of the energy spectrum). Thus, large-scale hemispheric-wide correlations are associated with variance in the $m = 0$ mode and, in spite of the fact that zonal jets are naturally produced by eddies on the sphere or β -plane, the covariability of flow around a circle of latitude may be relatively weak. Ultimately then, the importance of an annular mode is related to how much eddy energy is in the zonal modes, and this is a quantitative issue that can ultimately be settled only by an appeal to observations. The spatial structure of these correlations are in fact very similar to those found in various simulations with a general circulation model (Cash et al. 2002). There too the dipole structure of the EOF is apparent, but there is little hemispheric-wide correlation.

We can obtain another sense of the hemispheric vs local nature of the variability by constructing the EOFs from a quadrant (i.e., regions 90° wide) rather than the full hemisphere. In both cases the fields are put through a 10-day running average before computing the EOFs, and these are illustrated in fig. 8. The first EOF of the regional field is almost coincident with that constructed from the full hemispheric field, consistent with the notion that it is the same mechanism producing the variations in the zonal velocity on a hemispheric and on a regional scale. However, these variations are not always in concert. To quantify this, we compute the correlations between the daily timeseries of the principal components corresponding to the regional and hemispheric EOFs, and between two opposing quadrants (c.f., ?). The values of these are as follows:

$$C(Z, Q1) = 0.62 \tag{3.1a}$$

$$C(Z, Q2) = 0.63 \tag{3.1b}$$

$$C(Q1, Q2) = 0.16 \tag{3.1c}$$

Here $C(Z, Q1)$ is the temporal correlation between the first EOFs from the zonally averaged flow and flow in a quadrant (and similarly for $C(Z, R2)$), and $C(Q1, Q2)$ is the correlation between the flow in the two quadrants. The difference between $C(Z, Q1)$ and $C(Z, Q2)$ is solely due to the finite length of the timeseries, and so is a measure of the error due to that.

If there were a pure annular mode in the sense that the zonal velocity varied in unison on a hemispheric scale the correlations would all be unity. If the quadrants were completely independent

we would have

$$C(Z, Q1) = 0.5 \quad (3.2a)$$

$$C(Z, Q2) = 0.5 \quad (3.2b)$$

$$C(Q1, Q2) = 0. \quad (3.2c)$$

Clearly the flow here is something in between these extremes. One may conclude that although similar dynamics is acting on both the regional and hemispheric scale (because the meridional structure of the respective EOFs are so similar) this dynamics does not necessarily act in unison. We cannot expect the real atmosphere to have quantitatively the same values as (3.1), but the qualitative picture is likely to be similar.

c. Low and high index states

As noted, the mean position of the jet is slightly poleward of the center of the stirring. The stronger the jet the more poleward the mean jet position, as indicated in fig. 9, although the effect is rather weak and the displacement of the jet is no more than 5° . However, a stronger jet is also noticeably narrower than a weak one, and the easterlies on its equatorial flank are noticeably stronger and extend further poleward. The effect of this would be to enhance the perceived separation between the eddy driven jet and the subtropical jet, and to make the midlatitude surface westerlies both stronger and slightly more poleward during high index states, both of which effects are seen in the observations (Ambaum et al. 2001).

4. Zonally asymmetric model

a. One enhanced stirring region - the NAO

Suppose we now enhance the stirring in a longitudinal region in order to roughly mimic the effects of a storm track. However, we keep the simple meridional structure of a single stirring region in midlatitude, and the stirring maximum is at the same latitude for all longitudes. Specifically, the longitudinal structure of the amplitude of the stirring is

$$|F_\zeta| = A \left(1 + B \exp(-(x - x_0)^2 / 2\sigma^2) \right) \quad (4.1)$$

where A and B are constants. A determines the strength of the uniform background stirring and B that of the zonal inhomogeneity, centered around longitude x_0 . We have conducted experiments with B ranging from 0 to about 10, with a value of order unity perhaps best representing the enhanced stirring of the stormtrack regions over the Atlantic and Pacific. The parameter σ determines the width of the enhanced stirring region. Again we have conducted a number of experiments; an enhanced stirring of about 45° extent is roughly representative of the Atlantic stormtrack.

Fig. 10 shows the fields of eddy kinetic energy and the first EOF in an integration with $B = 1$. The eddy kinetic energy is a direct reflection of the enhanced stirring and, clearly, the EOF is centered around the enhanced stirring and reflects the more vigorous wobbling and pulsing in that region. The localized dipole structure of the streamfunction is very similar to that appearing in zonally asymmetric GCMs (e.g., Cash et al. 2002) and in the observations (e.g., Ambaum et al. 2001). The one point correlation function is calculated for this case, with a basepoint at the longitude where the EOF is a maximum. The correlation function picks up the meridional dipole structure of the EOF, just as in the zonally symmetric case. In the zonal direction, the correlation function is somewhat more localized than the EOF and is not, in fact, very dissimilar from that in the zonally symmetric case. The day-to-day synoptic activity in the two cases is rather similar, but in the zonally asymmetric case there is a slight preference for dipole structures to form in the region of enhanced stirring, and this is detected by the EOF analysis. In the zonally symmetric case, similar two-dimensional structures form, but with no longitudinal preference and the first EOF is almost zonally uniform.

The structural similarity between the EOF and the teleconnection, and the similarity between the barotropic model, the GCM, and the observations, are all suggestive of the robustness of the mechanism identified. We emphasize two points. First, this is a *nonlinear* effect. If localized stirring is added to the linear barotropic vorticity equation, then the response is a superposition of beta plumes that trail westward from the source but which produce no vorticity flux, an effect familiar to most physical oceanographers. Second, the dipole structure that is so reminiscent of the NAO arises fairly robustly when the stirring is somewhat stronger than the zonal mean stirring (i.e., when B in (4.1) is of order one). However, if the localized stirring is extremely intense then more exotic patterns (not shown) occur. Now the theory of section 2(a) becomes invalid because of the extreme zonal inhomogeneity.

b. Two stirring regions

The Northern hemisphere has two major storm tracks, one over the Pacific and the other over the Atlantic. We model this with two stirring regions, uncorrelated from each other, and chosen to give an eddy kinetic energy pattern that roughly corresponds to that observed. The eddy kinetic energy and the first EOF of streamfunction are illustrated in fig. 12, and the correlation function in fig. ???. The first EOF of the hemispheric field is very annular, with two fairly weak centers near the stirring regions. The correlation functions at these centers of action are rather similar to the case with only one storm track. The correlations between the two centers are rather weak, increasing with the latitude of the basepoint, and there is no pronounced teleconnection between the two stirring regions.

5. Temporal Structure

Apart from the signals due to El Niño and the seasonal cycle, the large-scale patterns of extratropical variability in the atmosphere appear to have a fairly red spectrum, with no really significant peaks (Feldstein 2000). However, it is unclear whether the power in these patterns continues to increase for timescales longer than the interannual — that is, whether the spectrum continues to redden for increasingly long timescales or whether it flattens out and whitens (Stephenson et al. 2000). Notwithstanding that uncertainty, the decorrelation timescale associated with the NAO and similar patterns is of order 10 days. In our model the various timescales are the timescale of the forcing, a frictional timescale determined by the value of r in (2.16), a nonlinear eddy turnover time for some scale L given by $L/|U_L|$ where U_L is the velocity magnitude at the scale L , and a timescale associated with Rossby wave propagation $\sim 1/(L\beta)$. The external parameters are those associated with the forcing and friction — the eddy turnover time is ultimately given by the magnitude of the forcing and how effective it is in generating flow. Typically, the shorter the decorrelation timescale of the forcing the less effective it will be in generating flow, as the response to high frequencies is out of phase with the forcing — the phase of the forcing changes too rapidly for it to have an effect on the vorticity. If we choose our forcing decorrelation time to be of order a few days to represent baroclinic activity, then we must tune its magnitude to give flow velocities with a magnitude similar to those observed, and in that case the only remaining free parameter is r and β .

Friction is clearly an important element in the reddening of the forcing spectrum, as we see from

the linear version of (2.16). The equation is

$$\frac{\partial \zeta}{\partial t} + \beta \frac{\partial \psi}{\partial x} = S - r\zeta \quad (5.1)$$

and this can be solved analytically if the power spectrum of S is known, assuming a solution of the form

$$\psi = \text{Re} \Psi e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}. \quad (5.2)$$

Substituting into (5.1) gives

$$|\Psi|^2 = \frac{S_\omega^2}{[(k^2 \omega^2 + \beta k_x)^2 + r^2 k^4]} \quad (5.3)$$

and so the forcing spectrum is reddened. To this should be added the solution of the homogeneous problem, a decaying Rossby wave.

Numerical solutions of the nonlinear problem show this effect — in fig. 13 we see a representative timeseries of the first EOF in a zonally symmetric simulation. (The EOF itself is first obtained using temporally low-passed data, but the figure shows the daily, unfiltered, values of the corresponding principal component.) Fig. 14 show the corresponding power spectra, which are characteristically red, with more power at low frequencies when the friction is small. The auto-correlation of the first two EOFs is shown in fig. 15, and these are of order 10 days. The wobbling mode typically has a longer decorrelation than the pulse, but as noted previously these two modes are not wholly independent, because as the jet wobbles from one extreme latitude to another it passes through its mean location twice, and in that sense pulses twice for a single wobble. Typically, for shorter timescales the decorrelation timescale (e.g., the efolding timescale) is roughly proportional to the frictional timescale, as expected in an Ornstein-Uhlenbeck type process. Somewhat unexpectedly, the auto-correlation of the simulation with the higher frequency has a shoulder at about 40 days, and has virtually as much power at very long times as the simulation with 23 days, indicating that frictional effects are not the sole determinant of the power at low-frequencies. (The correlation and power spectra are quite robust, coming from a 100 year simulation.) Note that it is also apparent from the timeseries that quite long excursions from the mean are possible. For example, there is almost a 500 day excursion between 6000 and 7000 days, and frequent excursions of order 100 days — note for example the dip centered around 300 days. These may have a similar mechanism to the long time periods of James (xxx) ?, but we have not explored this here.

6. Summary and Conclusions

We have presented a simple dynamical model of the North Atlantic Oscillation and the related annular modes. We have shown that spatial structures similar to those associated with the North Atlantic Oscillation and annular modes can be robustly and easily reproduced with a stochastically forced, but nonlinear, nondivergent barotropic model.

The model suggests that the NAO and annular modes are, essentially, two sides of the same coin. The (single) phenomenon is associated with variations in the midlatitude eastward flow caused by variable stirring from baroclinic eddies — the stirring is variable, because the dynamics of baroclinic eddies is chaotic. The stirring produces a largely barotropic eastward flow, and the strength and position of this will vary as the stirring varies. These variations can be represented as a combination of wobbling (the jet changing its latitude) and pulsing (the jet changing its strength), although these should not be construed as physically independent modes. We showed that this naturally produces a dipolar structure in the streamfunction (i.e. the pressure field), much as is observed. This wave-meanflow interaction produces variability in the zonally averaged flow, and this may be interpreted or defined as an annular mode: note that if the eddy statistics are zonally uniform, the leading EOF will be zonally uniform. However, in this interpretation the variability of an annular mode is not the hemispheric-wide synchronous variability or heaving of a polar vortex. Rather, it is the projection onto the zonally averaged flow of eddy dynamics.

The North Atlantic Oscillation is to be differentiated from the annular mode primarily by its scale, not its mechanism. The presence of an Atlantic stormtrack provides stronger stirring, and if the longitudinal extent of the stormtrack is greater than that of a single eddy, the same dynamics that produce variations of the zonally averaged flow will still act, just more intensely, over that region. Thus, the jet variations will be stronger here than elsewhere, and any measure of that variability, such as the first EOF of pressure or streamfunction, will show a dipole centered near the eddy activity. Now, when the stirring is stronger — i.e., when the stormtrack is stronger — the barotropic jet is strengthened and tightened, whereas the subtropical jet is little altered. Thus, during periods of high eddy activity the eastward advection will be strongest at latitudes poleward of its mean position. Conversely, quiescent periods will have a weaker barotropic jet and the eastward advection will be somewhat equatorward of its mean position. Thus, at one extreme we can expect eddy-rich activity with a strong eastward jet somewhat poleward of its mean position; at the other extreme we

expect weaker eddy activity with a weaker, slightly more equatorward jet. This is, of course, the manifestation of the NAO.

Another way of expressing this is to say that it is the organization of the baroclinic activity into spatially coherent large scale patterns (i.e., storm tracks) that gives rise to coherent large scale vorticity stirring, and this in turn produces patterns like the NAO. Because the mean amplitude of the vorticity-stirring varies zonally, the eddy forcing has a stationary component (i.e., there is a zonally asymmetry in a time mean eddy fluxes) and it is this stationary component that produces the NAO. Momentum fluxes from stationary waves are really the same as the spatially nonuniform eddy fluxes we have parameterized, and that such forcing is responsible for the zonally asymmetric patterns of variability seems consistent with the observational analyses (Limpasuvan and Hartmann 2000).

If this mechanism is the case, then there should be a corresponding phenomena in the Pacific as well as the Atlantic corresponding to the Pacific stormtrack. Such a ‘North Pacific Oscillation’ may well exist (indeed Walker and Bliss 1932, commented on it) although it may not be as noticeable as the NAO both because the high frequency variability in the Pacific is less than in the Atlantic, and because there are many other phenomena occurring in the Pacific, such as ENSO, the Pacific North American pattern and the Pacific Decadal Oscillation. There may well be additional, more subtle differences in the stormtracks between these regions and we recognize that the differences between the NAO and NPO may not be fully explained by our proposed mechanism.

The decorrelation timescale of the NAO and annular modes are observed to be about 10 days, and this is well reproduced by the model, albeit it is partly dependent on the frictional timescale chosen. The eddy forcing itself, even the stationary-eddy forcing, has a much shorter decorrelation timescale (a day or two) and this is reddened by damping processes and, to some degree, nonlinear dynamics. The barotropic model does produce variability on long timescales, evidently up to 1000 days, although the presence of a seasonal cycle might affect this. Determining whether this long-term variability corresponds to that seen in the observations will require a more detailed study of both model and observations, since the nature of such long-term variability is currently unclear in both. [?] concludes that the NAO is a Markov process with an e-folding timescale of about 10 days, whereas Stephenson et al. (2000) note the presence of ‘long-range dependencies’ (a red spectrum) on interannual timescales. These may not be contradictory, if the tails in the autocorrelations are

nonzero but small.]

Perhaps what is notable is what is *not* in the model we have presented — for example there is no stratosphere and no ocean. Both the stratosphere and the ocean may of course contribute to the persistence and intensity of the NAO and annular modes, but this model suggests that neither are responsible for its basic existence. It does suggest that the ocean might influence the NAO by affecting baroclinic activity and the stormtrack, but perhaps has less to say about stratospheric effects. Another simplification is that baroclinic effects are modelled by a simple stirring. This ignores the possibility that the stirring may be dependent on the strength of the jet. This might then lead to eddies following the position of the jet, and thereby the production of a longer timescale. The mechanisms in this paper have little to say about the existence or importance of that mechanism. Careful studies with a trustworthy GCM are likely to be needed before one can definitively determine the importance of an actual feedback mechanism between eddies and zonal flow in producing intraseasonal variability, especially as the observations do not clearly discount a simple stochastic process.

Although the simplicity of the model and the robustness of the results are appealing, some aspects of large-scale variability have *not* been elucidated, and we list some of them here: (i) The precise dynamical mechanism of the wobbling and pulsing of the zonally averaged jet. A weakly nonlinear theory may help here. (ii) The theory of the production and variability of a *zonally varying* zonal wind by vorticity stirring (section a) is rather ad hoc, as wave-mean flow interaction theory for a zonally varying flow is not well developed (see for example Swanson (2002)). (iii) The origin of the organization of the storm tracks, or enhanced baroclinic activity, and its precise relationship to the pattern of planetary waves in the atmosphere has not been addressed (this is a topic of much research elsewhere). Rather, we took as an input to the model the fact that baroclinic activity is not zonally uniform in the Northern hemisphere. (iv) There seems to be a tendency for the maximum amplitude of the EOFs to be located a little downstream of the storm tracks. We might expect this because momentum fluxes occur primarily in the decaying phase of the baroclinic lifecycle, and thus downstream of the center of the stormtracks. Such a mechanism cannot be reproduced in a barotropic model, and nor, in fact, is it a robust feature of simulations with GCMs (Cash et al. 2002).

In spite of these shortcomings, we have presented a simple, dynamically robust mechanism that

reproduces some of the main spatial and temporal characteristics of the large-scale variability in the atmosphere. We hope it may be useful as a dynamical basis for more complete models and simulations, and in interpreting the observations.

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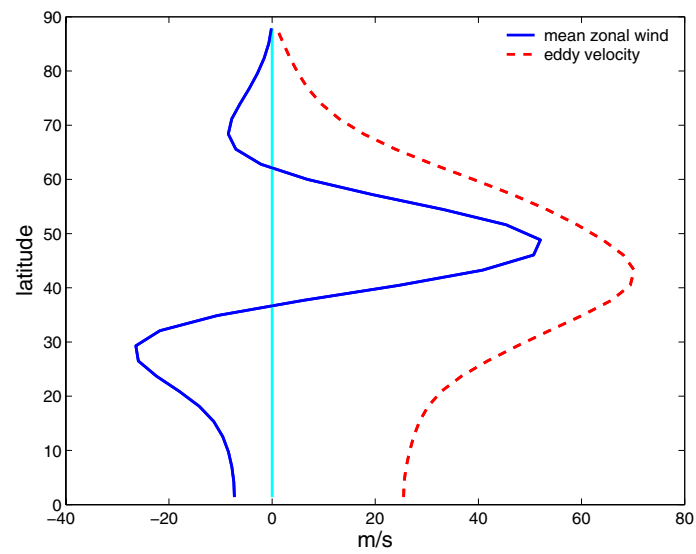


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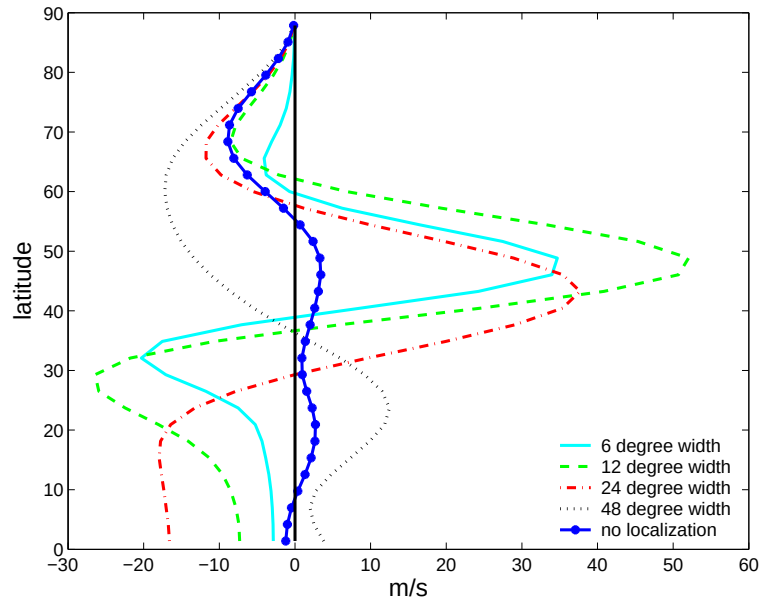


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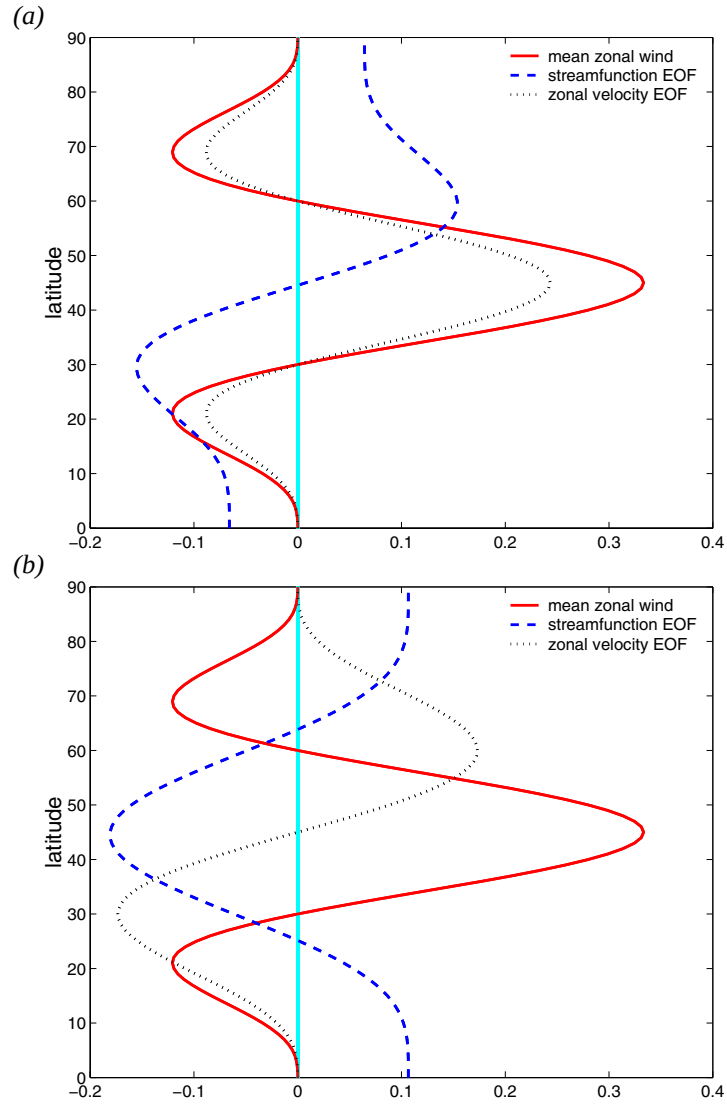


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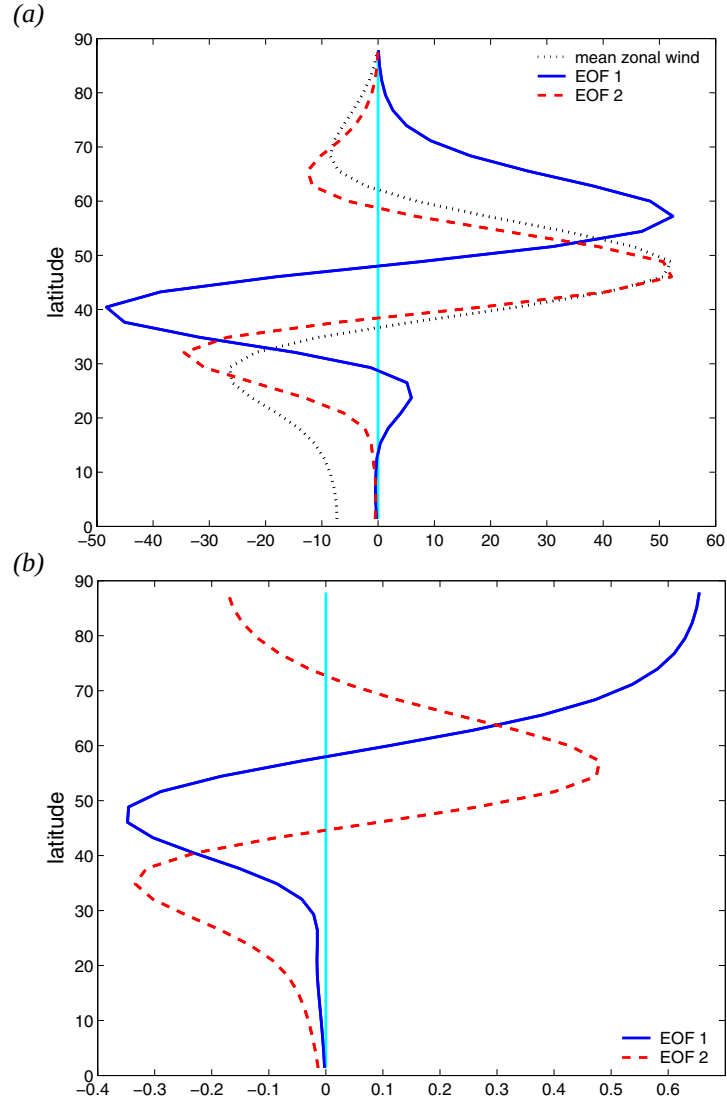


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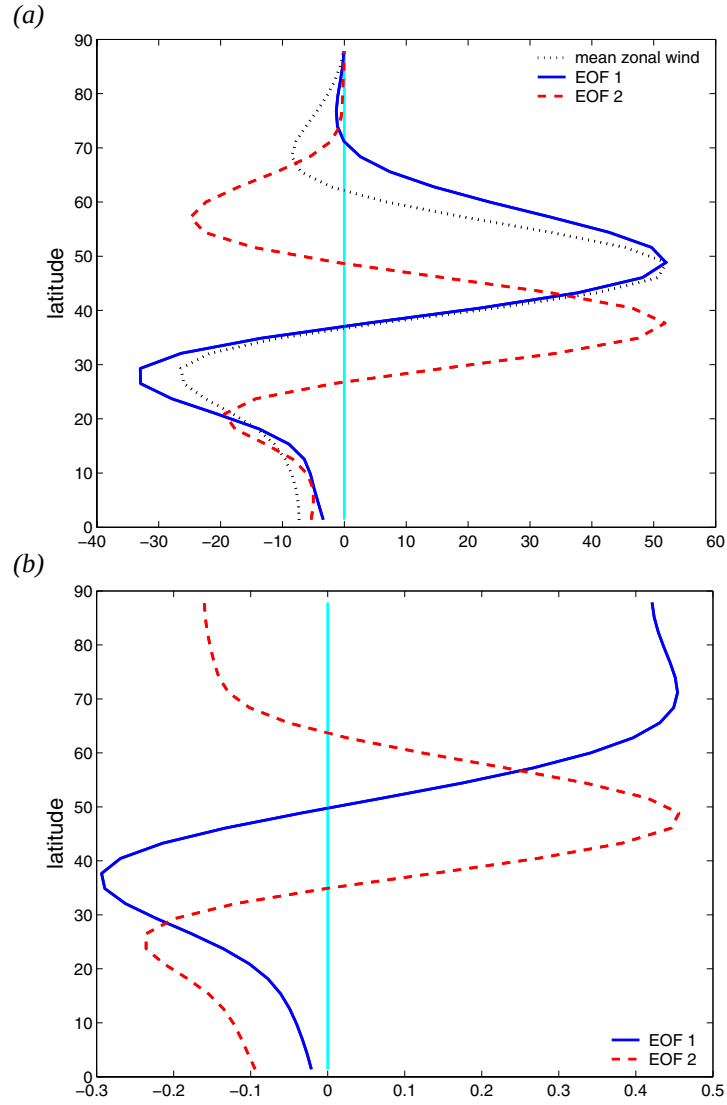


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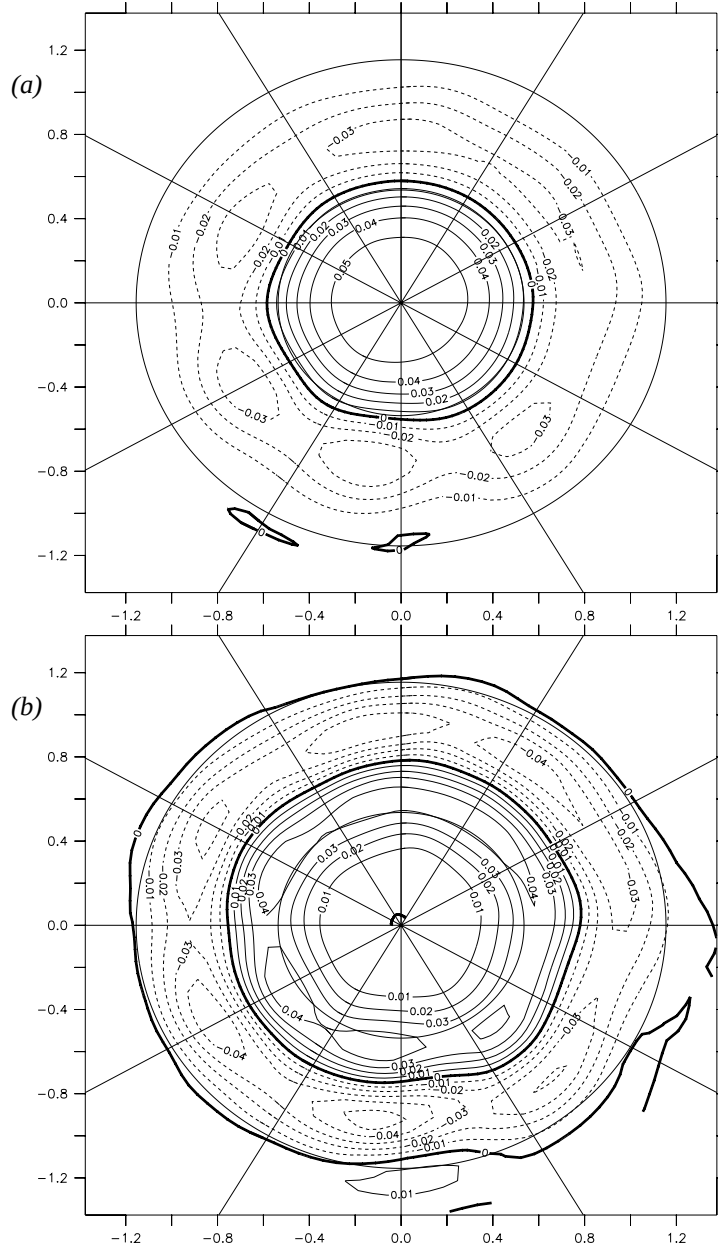


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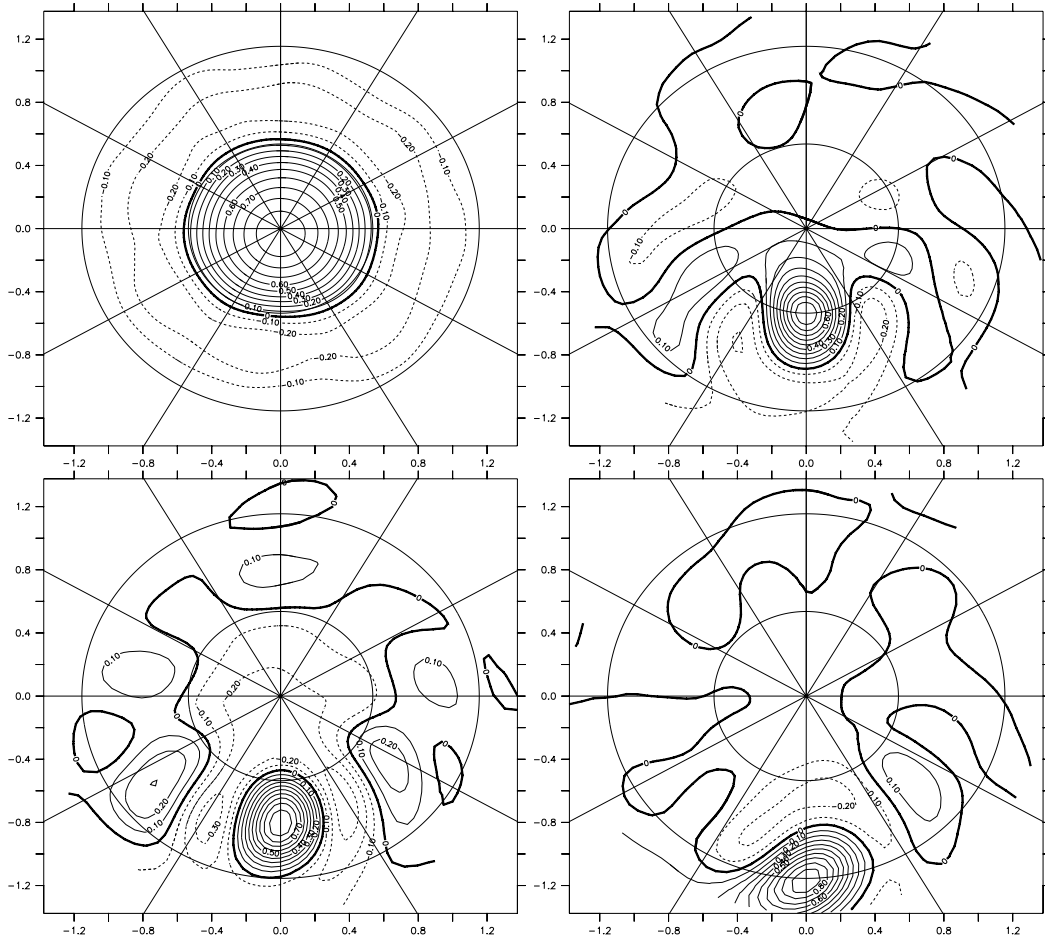


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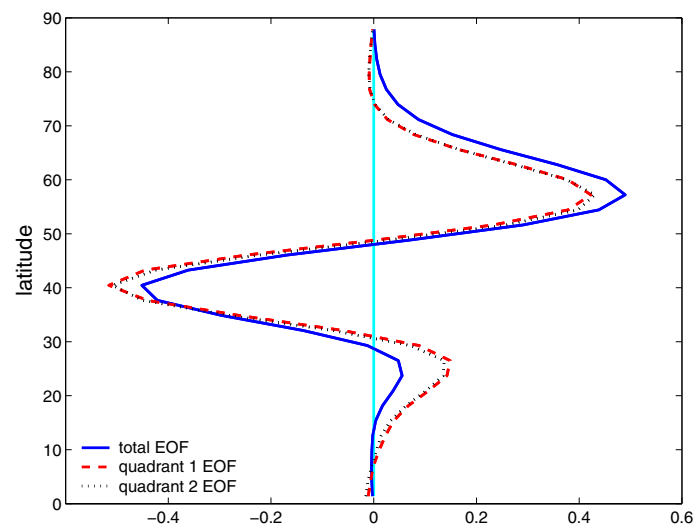


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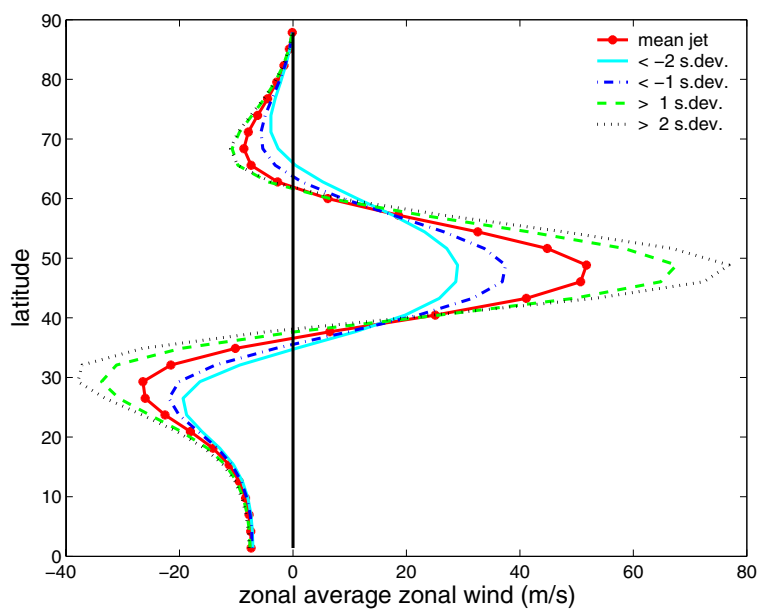


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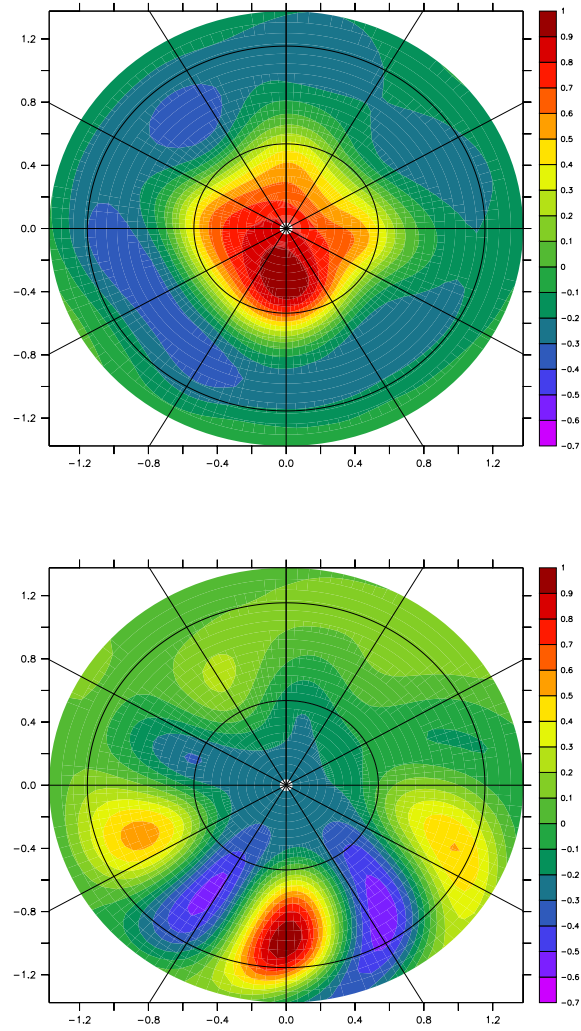


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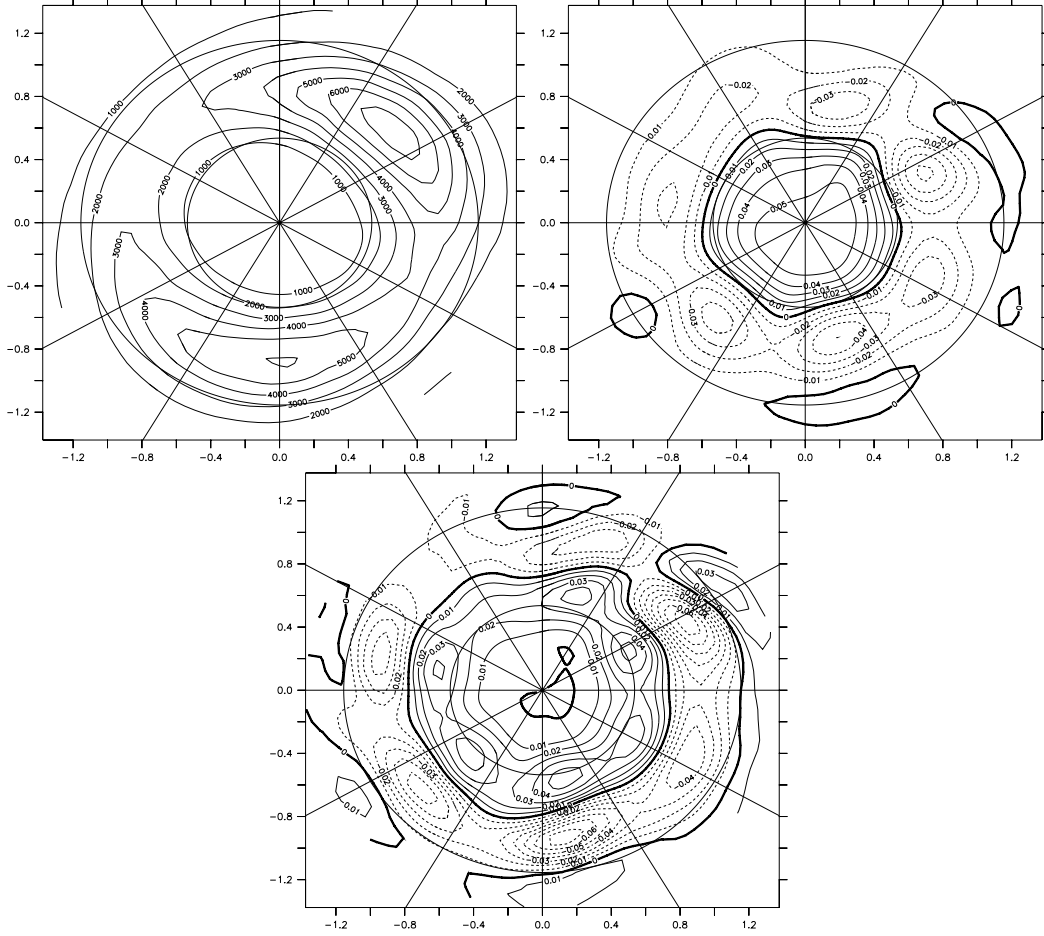


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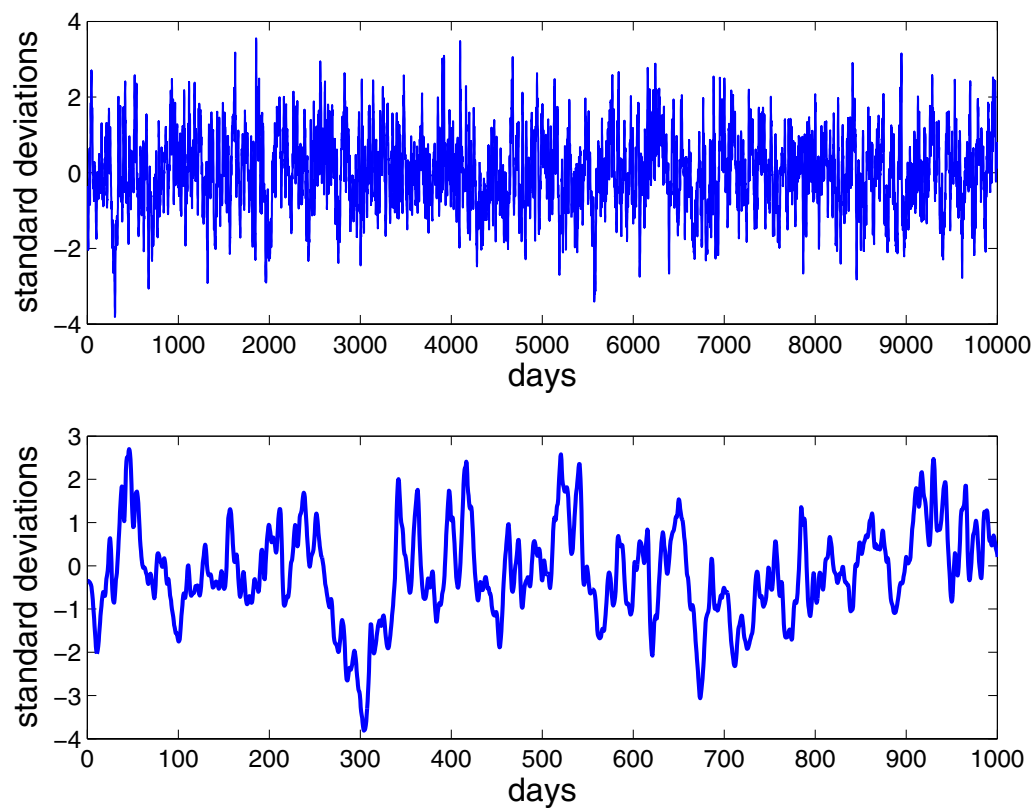


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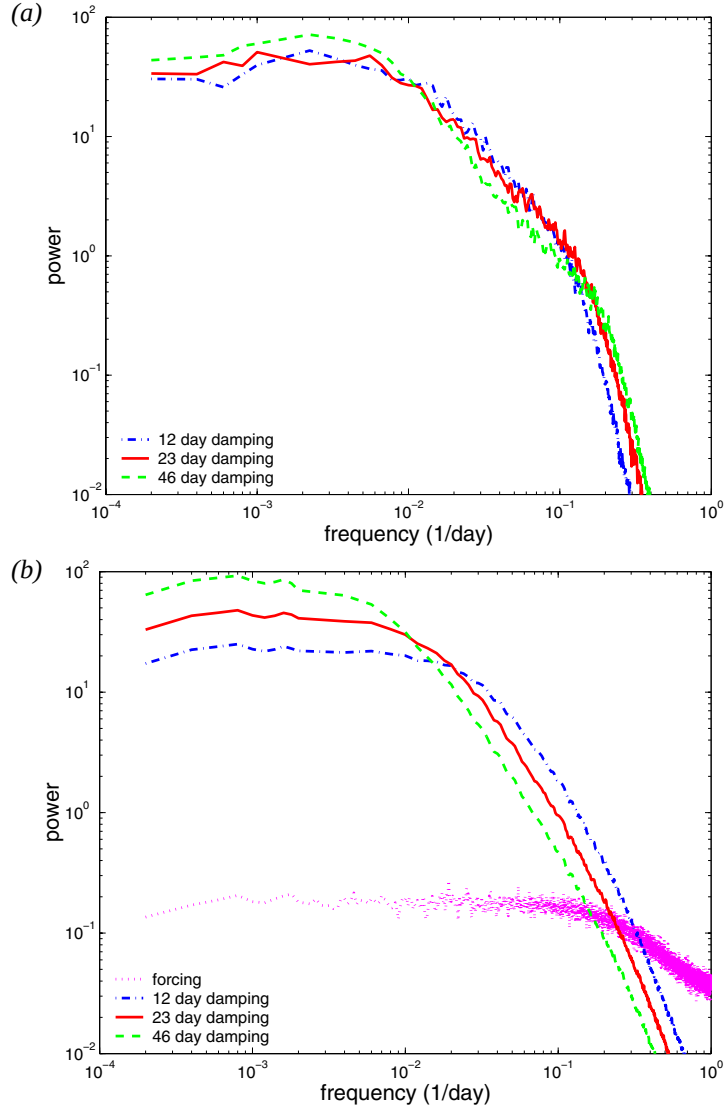


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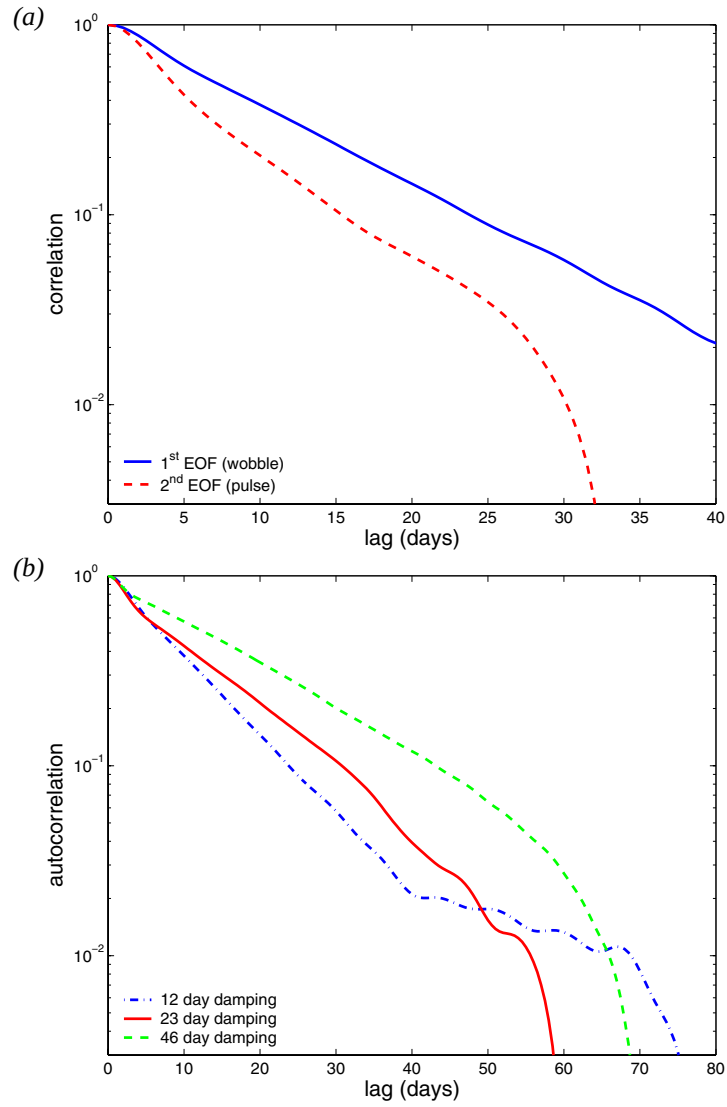


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